# Shuo REN

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**PhD researcher building long-horizon, tool-grounded LLM agents for complex engineering workflows.** Research spans multi-agent physical-design optimization, failure-conditioned repair, and automatically generated evaluation environments, grounded in verifiable EDA and mathematical-optimization tools.

## Internship & Education

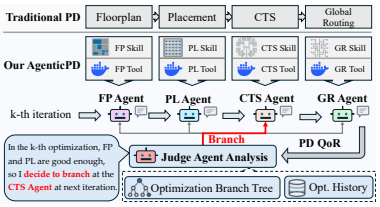
**Research Intern, EDA Research Lab, Hong Kong Science Park** May 2026–Present
Developing agentic 3D-IC workflows that couple LLM agents with industrial physical-design tools and feedback for partitioning and design-space exploration.

**Research Collaboration, ABCO Project (CUHK Industry Collaboration)** Mar 2025–Apr 2026
Built the physical-design-to-architecture feedback loop behind CLIP-3D, connecting analytical 3D floorplanning with gem5, McPAT, CACTI, and HotSpot.

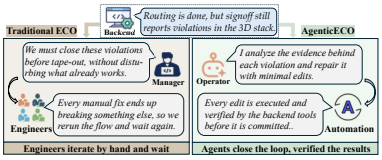
**The Chinese University of Hong Kong (CUHK)** Sep 2024–Present
PhD in CSE, full postgraduate studentship; working with Prof. Tsung-Yi Ho and Prof. Bei Yu

**Huazhong University of Science and Technology (HUST)** 2020–2024
BEng in Computer Science and Technology; GPA 3.93/4.00, top 1.5%; **National Scholarship of China**

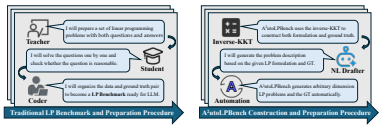
## Selected Agent Research




**AgenticPD** [link] [code] formulated chip physical-design optimization as a stage-aware tree search rather than flat parameter tuning or one-shot script generation. Built a hierarchical harness where a Judge Agent navigates reusable intermediate design states and stage agents act through stage-scoped tools with structured execution feedback and context management; evaluated across six design/technology combinations.




**AgenticECO** [link] [code] formulated post-route ECO for 3D-IC chip design as failure-conditioned agentic repair. Typed timing, routing, layout, and violation evidence drives deterministic repair programs executed outside the LLM. Every edit must pass immutable DRC, timing, structural-equivalence, and preservation gates; the closed loop cleared 7/9 matched cases versus 2/9 for stock repair and 9/9 in a cross-backbone rerun.




**A$^2$utoLPBench** [link] [code] replaced fixed LP corpora with an inverse-KKT generator and an agent-runnable Docker environment producing unlimited problems with certified optima and tunable difficulty. Across five frontier models, the solver–critic loop achieved a best gain of 15.1 percentage points on MAMO Complex, with repeatable trajectories and audited token costs.

## EDA Research Background

**Partitioning-Free 3D-IC Floorplanning** [paper], removed up-front netlist partitioning and jointly optimized block placement and die assignment under physical constraints; accepted at ASP-DAC 2026.

**CLIP-3D** [link], connected gem5, McPAT, CACTI, HotSpot, and analytical 3D floorplanning to expose thermal, wire, and cache effects during architecture exploration; improved realized BIPS by 13.75% on average across thermally binding configurations; featured as an Agentic 3D-IC case study in an IDAS 2026 keynote.

**Chiplet3D** [link], embedded exact pin transformations and a convolutional 3D thermal field in a MILP floorplanner; reduced wirelength by 39–43% on average and peak temperature by up to 45.9°C versus prior baselines.

## Technical Stack

**Agent/LLM**: Python; tool-use and multi-agent systems; structured tool APIs; Docker evaluation harnesses; trajectory, failure, and token-cost analysis; AI-native development with Claude Code and Codex.

**EDA/Systems**: C/C++, Linux, Git; OpenROAD/ORFS and Innovus; physical design, STA, and 3D-IC workflows; gem5, McPAT, CACTI, and HotSpot.